

Lecture 9: Nudges and Choice Architecture

Designing Better Choices Without Restricting Freedom

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ECON 230 · Introduction to Behavioural Economics · Winter 2027

The Cafeteria Experiment

A Simple Intervention with a Dramatic Result

A school cafeteria manager noticed children ate far more of whatever food was **placed first** in the line. She moved:

- Fruits and salads → **front** of the line
- Desserts and fried foods → **last**

Result: Fruit consumption rose **35%**. Fried food consumption fell **30%**.

No prices changed. No menu items removed. Just order.

The Key Insight

There is **no neutral choice architecture**. Whether you realise it or not, every choice environment has been designed — by someone, somehow. The only question is whether it is designed *thoughtfully*. This is the foundation of nudge theory.

1. **Libertarian paternalism** — the philosophy of nudges
2. Nudge vs mandate vs market
3. Default effects — the most powerful nudge (organ donation, pensions)
4. Framing effects — gain vs loss, attribute framing, decoy effect
5. Simplification and salience — reducing friction, making costs visible
6. Social norms as nudges — descriptive and injunctive norms
7. EAST Framework — designing effective nudges
8. Government nudge units — the UK BIT, Canadian evidence
9. Applications: health, environment, finance, tax compliance

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What Is a Nudge? (Thaler & Sunstein, 2008)

Definition

A **nudge** is any aspect of the choice architecture that alters people's behaviour in a predictable way **without forbidding any options** or significantly changing the economic incentives.

Libertarian Paternalism

- **Libertarian:** All options remain freely available. Complete freedom preserved.
- **Paternalistic:** The design of the choice environment is intended to steer people toward what is *good for them* (as they would define it on reflection).

Intervention	Mechanism	Example	Freedom?
Market/Price	Financial incentive	Carbon tax	Partial
Nudge	Default/framing	Opt-out pension	Yes
Information	Disclosure	Calorie labels	Yes
Mandate	Legal requirement	Seatbelt law	No

What Makes a Good Nudge?

Thaler & Sunstein Criteria

1. **Transparent:** People can see that they are being nudged and understand why.
2. **Easy to opt out:** Switching from the default is simple and costless.
3. **Evidence-based:** Grounded in robust knowledge of actual behaviour.
4. **Benefits the chooser:** Not designed to exploit the chooser for the nudger's gain.
5. **Accountable:** The nudge can be evaluated, modified, or reversed.

Counter-Example: Predatory Nudges

- Pre-checked boxes for insurance add-ons on travel booking sites.
- Auto-renewing subscriptions with hard-to-find cancellation.
- Casino floor layouts designed to disorient and keep you gambling.

Thaler & Sunstein (2021) call these “**sludges**”: bureaucratic obstacles that prevent people from accessing services or exercising rights. *Sludge should be eliminated.*

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Group Design Exercise — 4 Minutes

Group A: UNBC students don't exercise enough. Design 3 nudges.

Group B: UNBC students waste 30% of cafeteria food. Design 3 nudges.

Group C: Only 15% of UNBC students got the flu shot last year. Design 3 nudges.

Group D: Students arrive late to class 40% of the time. Design 3 nudges.

Evaluation Framework

- Is it libertarian (can people opt out)?
- Is it paternalistic in a legitimate direction?
- Does it exploit a specific bias from our lectures?
- What is the evidence that this type of nudge works?

The Power of Defaults

Why Defaults Are the Most Powerful Nudge

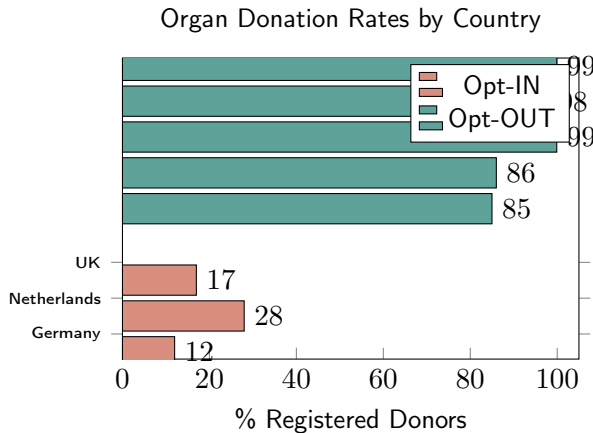
A **default** is what happens if you do nothing. Due to inertia and present bias, people tend to stick with defaults. Three mechanisms:

1. **Inertia/procrastination:** Switching requires effort; effort requires overcoming present bias.
2. **Implicit endorsement:** “The government/company set this default — it must be the right choice.”
3. **Loss aversion:** Switching from the default feels like giving something up.

The Apple Privacy Example

Apple changed the default for iPhone ad tracking from **opt-out** to **opt-in** (iOS 14, 2021). Result: only **21%** of users opted into tracking. Meta (Facebook) lost an estimated **\$10 billion** in annual revenue from one default change. Source: Snap investor call, 2022.

Organ Donation — The Canonical Example



Same Population, Different Default

Austria (opt-out): **99.9%** registration.

Germany (opt-in): **12%** registration.

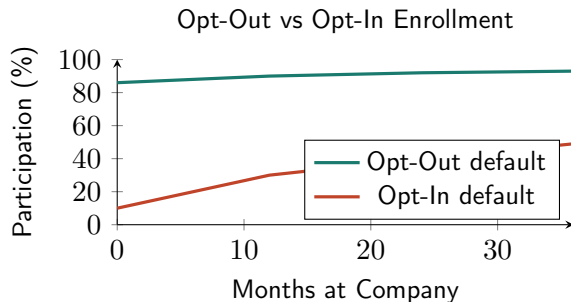
These countries are culturally similar. The difference is *entirely* due to the default.
(Johnson & Goldstein, 2003)

Canadian Context

- Canada: opt-in; **32%** registered. **1,600** die on waitlists/year.
- **Nova Scotia**: opt-out (2021). Registration rising.
- Ontario: prompted choice at licence renewal $\Rightarrow +6$ pp.

Pension Auto-Enrollment

Madrian & Shea (2001): Changed 401(k) from opt-IN to opt-OUT. Same plan, same rates, different default.



The Gap

- **Opt-in after 3 years:** 49%
- **Opt-out immediately:** 86%
- **Gap:** 37 percentage points
- *from one word change in an HR document*

International Adoption

- **US:** Pension Protection Act (2006) — auto-enrollment nationwide.
- **UK:** NEST — 90%+ participation.
- **Australia:** 11% mandatory.
- **Canada:** Federal pilot (2019) — 40% participation increase.

Gain vs Loss Framing in Policy

Rothman & Salovey (1997): **Detection** (mammogram) $\Rightarrow$ **GAIN frame** (“Early detection saves lives.”). **Prevention** (vaccine, sunscreen) $\Rightarrow$ **LOSS frame** (“Risk cancer without sunscreen.”).

Canadian Carbon Levy

Renamed “Climate Action Incentive Payment” — framed as a rebate (gain) not a tax. Political support significantly higher. BC carbon tax (2008): revenue-neutral framing $\Rightarrow$ one of the most popular carbon taxes globally.

The Decoy Effect (Ariely, 2008)

Economist: online \$59 / print \$125 / print+online \$125. Print-only exists only to make print+online a bargain. Without decoy: 68% chose online. With decoy: **84%** chose print+online.

Cialdini (2006) — Two Types of Social Norms

- **Descriptive norm:** “Most people do X.” Describes what *is* common.
- **Injunctive norm:** “People *should* do X.” Describes what is *approved* of.

Both influence behaviour. But descriptive norms can **backfire** (boomerang effect).

Opower Energy Experiment (Allcott, 2011)

Utility companies sent “Home Energy Reports” comparing household usage to similar neighbours.

- Average reduction: **2%** — 6M customers $\times$ 2% = savings of 1 power plant.
- Cost per kWh saved: **\$0.02** vs \$0.05 to build new capacity.
- **BC Hydro “Team Power Smart”:** 10%+ energy reduction (100,000 participants).
- **Boomerang effect:** Below-average users *increased* usage. Fix: add a happy-face emoticon ($\approx$ 0.3% additional reduction) = \$3M annual savings.

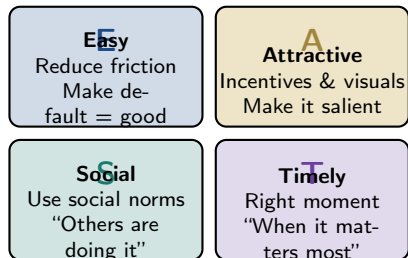
The Behavioural Insights Team (UK) Experiment

Added one sentence to UK tax reminder letters: *“9 out of 10 people in your area pay their taxes on time.”*

- Late payment rate fell **15%**.
- Multiplied across millions of letters: tens of millions of pounds in additional annual revenue.

Canada Revenue Agency — Social Norms Trial (2019–2021)

- Tested norm message on 120,000 late filers: *“Last year, 93% of Canadians paid their taxes on time.”*
- Result: **10% faster filing** in pilot areas (CRA pilot results, 2021).
- Cost of sending the letter: near zero.
- **This is the highest ROI government intervention documented in Canadian public administration.**



BIT (Behavioural Insights Team)

Founded in UK Cabinet Office, 2010. Now in 31 countries. Runs 750+ RCTs per year across government services. "The Nudge Unit."

EAST in Action

- **Easy:** Pre-fill OSAP with CRA data.
- **Attractive:** Colourful energy efficiency labels.
- **Social:** "Your neighbours recycled 80% this month."
- **Timely:** Text vaccine reminder 3 days before your appointment.

Impact Canada — Behavioural Insights Team (2018–):

- **Veterans:** Simplified application $\Rightarrow$ +40% uptake.
- **Employment Insurance:** Clear letters $\Rightarrow$ +20% faster resolution.
- **Cancer Society:** Text reminders $\Rightarrow$ +30% screening bookings.
- **CRA tax:** Social norm message $\Rightarrow$ -10% late filing.
- **Canada Learning Bond:** One-click RESP $\Rightarrow$ 15% $\rightarrow$ **52%** takeup.

Principles

1. All interventions require RCT evidence.
2. Transparent; evaluated for persistence; pre-registered.

Group Exercise — Write the Best Norm Message

Each group writes a **one-sentence norm message** to encourage UNBC students:

- **Group A:** Return library books on time.
- **Group B:** Use reusable cups at Tim Hortons.
- **Group C:** Get vaccinated before fall semester.
- **Group D:** Attend professor office hours.

Evaluation

Class votes on most effective message. After voting, discuss:

- Does it use a descriptive norm (“most students...”) or injunctive (“students should...”)?
- Is there a risk of boomerang? How would you prevent it?
- Is it timely and salient?
- What evidence type (RCT, field experiment, survey) would you use to test it?

Health

- **Cafeteria design:** fruit at eye level $\Rightarrow$ 35% more consumed.
- **Plate size:** smaller plates $\Rightarrow$ 30% less food served.
- **Vaccine reminders + implementation intention:** “Your appointment: Monday 2pm, Clinic X” $\Rightarrow$ 15% uptake increase (Milkman et al., 2021).
- **Stair prompts:** footprints on stairs $\Rightarrow$ +25% stair use.

Environment

- **Green energy default** (UK, Germany): 7% opt-in vs 69% opt-out — same energy, different default.
- **Vancouver plastic bag charge** (2021): 80% reduction in single-use bags.
- **Carbon footprint on receipts** (Doconomy/Mastercard): CO₂ per purchase visible at checkout.
- **Real-time dashboards:** Nest/Ecobee smart thermostats show live energy cost.

Financial Nudges: What the Evidence Shows

- **TFSA default to balanced ETF** (not cash): TD pilot 2020 $\Rightarrow$ **35% more invested**. Inertia works in your favour once the default is changed.
- **Credit card minimum payment warning** (FCAC 2021): Disclosing “you will pay off this balance in X years” $\Rightarrow$ **12% reduction** in minimum-only payments.
- **Overdraft opt-in** (FCPB/credit unions 2018): Changing overdraft protection from automatic to opt-in $\Rightarrow$ **30% reduction** in overdraft fees among frequent users.
- **RESP one-click enrolment** (Canada Learning Bond via CRA MyAccount): Takeup from 15% to 52% in pilot — just by reducing friction.

The Common Thread

Every successful financial nudge either **changes the default**, **reduces friction**, or **makes long-run costs salient**. None restricts choice.

Discussion Questions

- Q1. “Is it manipulative for governments to use defaults to increase organ donation? What if *you* didn’t want to donate and forgot to opt out?”
- Q2. “Should corporations be *banned* from using nudges against consumers? Or is this just marketing — and where is the line?”
- Q3. “Design one nudge for UNBC that could improve either student health, academic outcomes, or sustainability. What evidence would you need before scaling it?”

Key Takeaways — Lecture 9

- ✓ **Choice architecture is unavoidable** — the question is only whether it is designed thoughtfully.
- ✓ **Nudges:** libertarian paternalism — preserve freedom while steering toward better outcomes.
- ✓ **Defaults:** most powerful nudge. Organ donation (99.9% vs 12%), pension (86% vs 49%).
- ✓ **Framing:** same information, different presentation $\Rightarrow$ different choices.
- ✓ **Social norms:** descriptive + injunctive norms shift behaviour; beware boomerang.
- ✓ **EAST Framework:** Easy, Attractive, Social, Timely.
- ✓ **Canadian nudge policy:** Impact Canada, CRA, FCAC — real, evidence-based, measurable.

Next: Lecture 10: Applications & Policy Design — Bringing it all together. When should we use nudges vs taxes vs mandates? How do we evaluate behavioural policies? And what are their limits?